Kumari Priya Lala

ms25mtech14006@iith.ac.in | priyalala2003@gmail.com | +91 9931591499 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.78
KENDRIYA VIDYALAYA BAILEY ROAD PATNA BIHAR	Class XII (CBSE)	2021	93.60%
KENDRIYA VIDYALAYA BAILEY ROAD PATNA BIHAR	Class X (CBSE)	2019	95.60%

WORK EXPERIENCE

Power Grid Corporation of India Limited (PGCIL) (Project Intern)| Link

- Worked on-site alongside PGCIL officials on the project "Communication in Electrical Transmission Systems and Telecommunication," gaining direct exposure to integrating power grid infrastructure with high-speed optical networks.
- Configured DWDM and CWDM multiplexing equipment and made physical fiber-optic connections on-site to establish high-capacity transport links across transmission nodes.
- Observed and diagnosed real-world transmission faults using EDFA, DCM, OTDR, and optical reflectometers, performing fiber link calibration and signal amplification checks under engineer guidance.
- Managed operational workflows within the SAP ERP portal and carried out secure remote device management and troubleshooting via Anydesk.

PROJECTS

Engineering Photocathode Materials for Efficient Hydrogen Generation | M-Tech Thesis

- Developing earth-abundant p-type Cuprous Oxide (Cu2O) photocathodes to drive the Hydrogen Evolution Reaction (HER) for photoelectrochemical (PEC) solar water splitting.
- Synthesizing 1D Cu2O nanostructures using Anodization, Electrodeposition, and RF/DC Sputtering, followed by Thermal Annealing to optimize crystallinity, film stoichiometry, and carrier transport.
- Executing material characterization using Scanning Electron Microscopy (SEM) for surface morphology and X-Ray Diffraction (XRD) for crystal structure and phase purity analysis.
- Passivating the semiconductor-electrolyte interface against photocorrosion using thin protective overlayers and integrated surface co-catalysts to accelerate HER kinetics and reduce charge recombination.

Microstructure Classification via PCA/SVD | Machine Learning & Data Analytics in Materials Science Course| Link

(Python, NumPy, scikit-learn)

- Simulated Cahn-Hilliard and Allen-Cahn phase-field microstructures; showed scalar descriptors (area fraction, interface density) failed to distinguish the two families
- Applied PCA/SVD on flattened microstructure images to separate families in latent space, despite top 2 PCs capturing only 15.8% of variance
- Built a logistic regression classifier with leave-one-out CV, improving accuracy from 56% (2 PCs) to 81% (9 PCs).

Computational Modelling of Atomic & Molecular Interactions | Scientific Computing Course|Link

- Developed Real-Time Simulation Engine using Python simulator utilizing custom numerical integration algorithms to compute multi-particle trajectories and force fields in real time.
- Implemented Physics & Computational Algorithms for non-bonded interatomic potentials, dynamic collision mechanics, and physical energy conservation (kinetic vs. potential) in confined molecular systems.
- Integrated high-performance data visualization libraries to stream live physical outputs, enabling interactive parameter adjustments and real-time monitoring of system thermodynamics.

Design & Optimization of Distributed Laser Charging-Based Wireless Power Transfer Systems | *B.Tech Thesis* |Link

- Designed a laser-based wireless power transfer system using an AlGaAs gain medium operating in the near-infrared

region (880 nm).
- Modeled pulse dynamics (10 ns pulse width, 5 µJ energy) to optimize spot size, peak power (500 W), and transmission efficiency.
- Engineered laser cavity parameters -- mirror reflectivity and gain medium characteristics -- to achieve stable stimulated emission.
- Achieved ~16.8% end-to-end system efficiency through optimized feedback control and energy transfer mechanisms.

Active Noise Cancellation for Headphones using Adaptive LMS Filtering | MATLAB, Digital Signal Processing Course| Link

- Designed a signal processing unit for active noise-cancelling headphones by implementing the Least Mean Square (LMS) adaptive filtering algorithm from scratch in MATLAB.
- Modeled a voice signal corrupted by noise, generated a reference noise signal, and applied an adaptive filter to iteratively estimate and subtract the noise using a normalized weight-update rule.
- Validated the approach by plotting and comparing the original voice, noisy signal, and adaptive filter output, demonstrating effective noise suppression.

IoT Environment Monitoring System for Asthma Patients | ESP32, MQ-series Gas Sensor, Cloud Platform, Machine Learning| Link

- Engineered an end-to-end IoT environmental monitoring system on ESP32, integrating a gas/air-quality sensor array with a cloud-connected software stack.
- Established a continuous data pipeline linking hardware sensors with cloud platforms to aggregate and analyze Air Quality Index (AQI) parameters across multiple Indian cities.
- Implemented machine learning models on multi-city AQI datasets to map environmental triggers against potential health risks for asthma patients

Real-Time IoT Pulse Oximeter Development | Arduino, Optical Sensor, Touch Sensor, Display| Link

- Developed a real-time embedded health-monitoring device on Arduino to measure pulse rate and peripheral oxygen saturation (SpO2) using an optical pulse-oximetry sensor.
- Integrated a touch sensor for user input and a display module for real-time readout, alongside firmware for artifact filtering and noise reduction to improve reading accuracy.
- Deployed prototype units to a control group of peers and faculty to collect real-time physiological data and validate device ergonomics and usability.

RELEVANT COURSES

Artificial Intelligence and Machine Learning, Python,Computer Communication and Networks ,Embedded C , Information Theory and Coding, Java, Internet Of Things Domain Analyst , Robotics and Automation, Data Structure and Algorithm, Scientific Computing

TECHNICAL SKILLS

- **Machine Learning & Scientific Computing:** NumPy, scikit-learn, PCA/SVD, Machine Learning, Numerical Modeling, Scientific Computing, Data Visualization
- **Programming & HDL:** Python, C++, Java, Embedded C, Verilog HDL, Assembly, Data Structures & Algorithms
- **TCAD & EDA:** Synopsys Sentaurus TCAD, Cadence Virtuoso, NI Multisim, MATLAB, COMSOL, TCAD Device Modeling
- **Cloud & Platforms:** Google Cloud Infrastructure, AWS, SAP ERP
- **Embedded & IoT:** 8051 Microcontrollers, Arduino, ESP32, UART/RS232, GPIO, Timers, Hardware Interrupts, Sensor Interfacing, Internet of Things (IoT)

CERTIFICATIONS

- **Digital Design Hands-on | Link**
- **Infrastructure and Application Modernization of Google Cloud | Link**
- **User Centric Computing for Human Computer Interaction | Link**

- **Value Added Course on IOT | Link**
- **AWS Online Conference | Link**